

An Integrated Fraud Defense Framework for Card-Based Account Funding in Bangladesh's Mobile Financial Services: Addressing Social Engineering–Driven OTP Bypass Through Fraud Triangle, Routine Activity, and NIST Cybersecurity Principles

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Abstract

This systematic literature review aims to address the increasing incidence rate of social engineering-based OTP bypass fraud targeting card-based Account Funding Transactions within the Mobile Financial Services ecosystem of Bangladesh. This study proposes an integrated conceptual model by synthesizing the findings through three complementary frameworks: Fraud Triangle Theory, Routine Activity Theory, and the NIST Cybersecurity Framework. The PRISMA guidelines were followed in a systematic search across Scopus, Web of Science as well as IEEE Xplore for studies published during 2015-2025, culminating in synthesis from 28 relevant studies. Analysis was based on a multi-theoretical approach to understanding of offender motivations, criminal opportunities, and security control gaps. The synthesis revealed that fraudster motivation (FTT) is driven by economic pressure and low perceived risk; criminal opportunities (RAT) emerging from technically uninformed users with insufficient digital guardianship; and security controls (NIST CSF) are disproportionately focused on "Protect" functions while ignoring "Detect" and "Respond" capabilities. An integrated socio-technical approach is required to combat OTP bypass fraud. We propose a novel conceptual model that integrates motive deterrence (addressing FTT), target hardening with AI-powered guardianship (addressing RAT), and a comprehensive control lifecycle (implementing NIST CSF), which will provide stakeholders with an actionable framework for enhancing MFS security in Bangladesh.

Keywords: OTP Bypass; Social Engineering; Mobile Financial Services (MFS); Mobile Banking; Fraud Triangle Theory (FTT); Routine Activity Theory (RAT); NIST Cybersecurity Framework (NIST CSF); Financial Fraud; Fraud Detection; Fraud Mitigation; Integrated Fraud Defense Framework; Account Funding Transactions (AFT); Card-Based AFT; Cybersecurity Controls; Bangladesh.

1. Introduction

Mobile Financial Services (MFS) have been growing rapidly in Bangladesh, and as a result, financial inclusion has advanced significantly. millions of unbanked people have entered the formal economy. bKash, Nagad, and Rocket are some of the platforms that have changed how digital payments are made by processing billions of Taka every day (Hossain et al., 2022). This tremendous growth has also brought sophisticated cybercriminal operations into the picture, particularly social engineering attacks that aim to bypass One-Time Password (OTP) security for card-based Account Funding Transactions (AFT). Perpetrators use vishing and other manipulative tactics in these attacks to deceive users into revealing their OTPs so that fraudulent transactions can be authorized by the criminals (Hossain, 2022).

The scale of this problem is substantial; the Criminal Investigation Department (CID) of Bangladesh reported that BDT 75,000 crore (USD 8 billion) was laundered through Hundi operators using MFS in one year (Huda & Sony, 2025). This specific threat exists within a broader context of increasing cybercrime in Bangladesh where digital fraud and identity theft are becoming more common (Pervin, 2025). Current understanding about OTP bypass mechanisms is fragmented across disciplinary boundaries. Studies typically address either technical vulnerabilities or user education or broader cybercrime trends in isolation. They fail to capture the complex interplay between offender motivation, environmental opportunity, and technical control failures.

This review argues that integrated theoretical perspectives bridging criminology and cybersecurity are necessary for a comprehensive understanding. A systematic literature review is therefore undertaken to synthesize existing knowledge by applying Fraud Triangle Theory (FTT) to explain why fraud happens; Routine Activity Theory (RAT) to describe how criminal events converge in cyberspace; and NIST Cybersecurity Framework (NIST CSF) as a structured blueprint for defense mechanisms. This tripartite analysis is further contextualized within Bangladesh's regulatory environment, KYC challenges, and emerging technological solutions such as AI and blockchain (Dewi et al., 2024; Mondal & Sarkar, 2025; Sarker, 2025).

The main question that leads this study is: How can a combined FTT, RAT, and NIST CSF model offer a broad framework for comprehending and reducing social engineering-based OTP bypass fraud within the Bangladeshi MFS environment?

2. Methodology (PRISMA Approach)

This systematic literature review followed the PRISMA guidelines to maintain methodological rigor and transparency (Page et al., 2021).

2.1. Eligibility Criteria

Eligibility criteria included peer-reviewed journal articles, conference proceedings, and credible institutional reports published between 2015-2025 in English. Studies were required to focus on MFS security, social engineering, financial fraud, IT governance, electronic payment security, or related cybercrime trends in Bangladesh or similar developing contexts.

2.2. Information Sources

Extensive searches were conducted in four major databases: Scopus, Web of Science, IEEE Xplore, and the ACM Digital Library. Further articles were found through backward citation tracking and manual checking of reference lists from pertinent papers to reduce publication bias.

2.3. Search Strategy

The search strategy used a Boolean-based query that combined technical, behavioral, and contextual terms. The final search string was:

("OTP bypass" OR "social engineering") AND ("mobile financial service*" OR "MFS" OR "digital wallet") AND ("fraud" OR "security") AND ("Bangladesh*" OR "developing country") AND ("NIST*" OR "framework").

2.4. Study Selection

The initial search retrieved 287 records; after deduplication 210 titles and abstracts were screened for relevance. Sixty-eight (68) full-text articles were assessed for eligibility with 28 studies meeting all inclusion criteria that will be synthesized finally. The process of choosing followed the PRISMA rules closely and is shown in the PRISMA flow diagram (Appendix A). This diagram describes the steps of finding, checking, deciding if it meets eligibility requirements, and including studies.

2.5. Data Extraction

Data extraction utilized a standardized matrix that focused on findings related to fraudster motives (FTT), opportunity structures (RAT), security controls (NIST CSF), and contextual and Institutional factors.

- **Fraudster Motivation** – assessed through Fraud Triangle Theory (FTT) and operationalized by pressure, opportunity, and rationalization elements (Cressey, 1953).
- **Opportunity Structures** – assessed through Routine Activity Theory (RAT) and operationalized by motivated offenders, suitable targets, and absence of capable guardians (Cohen & Felson, 1979).
- **Security Controls** – assessed through the NIST Cybersecurity Framework (NIST CSF) and operationalized by its five core functions: Identify, Protect, Detect, Respond, and Recover (NIST, 2018).
- **Contextual and Institutional Factors** – such as regulatory oversight and enforcement regime in place for MFS operations in Bangladesh; KYC compliance levels among customers; public awareness regarding frauds perpetrated against MFS customers; digital literacy levels of different strata within the Bangladeshi MFS ecosystem.

2.6. Data Synthesis and Analysis

A narrative synthesis was used that followed a deductive coding scheme. The findings were organized and interpreted through the same three guiding frameworks applied earlier (FTT, RAT, and NIST CSF).

This triangulated approach helped the review to:

- ✓ Describe why fraud takes place (motivational and rationalization factors – FTT).
- ✓ Investigate how fraudulent acts are put into practice (opportunity and guardianship gaps – RAT).
- ✓ Evaluate what cybersecurity controls are present or absent (control lifecycle – NIST CSF).

The combination of these dimensions allowed for the development of an all-inclusive socio-technical framework to better understand and reduce social engineering-driven OTP bypass fraud in Bangladesh's mobile financial ecosystem.

3. Theoretical Background

3.1. Fraud Triangle Theory (FTT)

FTT, first introduced by Cressey in 1953, suggests that three elements must converge for fraud to happen: Pressure, Opportunity, and Rationalization. In applying this theory to MFS:

- ❖ Pressure relates to economic hardship or the allure of incentives from criminal organizations, which is highly relevant in the context of Bangladesh's economy (Pervin, 2025).
- ❖ Opportunity is created by systemic weaknesses in security, lack of user knowledge about digital protocols, and regulatory shortcomings regarding KYC and transaction monitoring (Huda & Sony, 2025).
- ❖ Rationalization happens when offenders think about fraud as a low-risk crime against large corporate entities with no feelings involved and that can be easily done through digital means.

3.2. Routine Activity Theory (RAT)

Cohen and Felson's (1979) RAT states that direct-contact predatory crime involve three elements: a Motivated Offender, a Suitable Target, and the Absence of a Capable Guardian. In the case of digital MFS environments, these can be defined as follows:

- ❖ The Motivated Offender is represented by fraudsters and criminal syndicates using advanced phishing scripts (Rahman, 2023).
- ❖ The Suitable Target is users with inadequate digital literacy who make up a large part of the Bangladesh MFS user base (Sultana et al., 2022).
- ❖ The Absence of a Capable Guardian means there is no real-time transaction monitoring, weak authentication systems, or alert mechanisms in place as noted in MFS and financial security studies (Khan & Ahmed, 2022; Mondal & Sarkar, 2025).

3.3. NIST Cybersecurity Framework (NIST CSF)

The NIST Cybersecurity Framework provides a policy-based structure for managing cybersecurity risks through five key functions: Identify, Protect, Detect, Respond, and Recover (NIST, 2018). The framework takes technical controls and fits them into a formalized lifecycle of risk management. It also serves as a standard against which the maturity of an organization's cybersecurity can be measured. Unfortunately, this aspect is significantly underdeveloped in Bangladesh's Mobile Financial Services sector (Hossain et al., 2022).

3.4. Integration of Frameworks

Bringing together these frameworks creates an effective analytical perspective: FTT reveals the basic reasons for fraud (the motivation), RAT describes the circumstances that make criminal events possible (the method), and NIST CSF offers a systematic way to implement defenses (what actions should be taken).

4. Findings

4.1. FTT Lens: Motivations and Drivers

There is consistent evidence that economic pressure and low perceived risks of prosecution are major motivating factors for offenders (Islam & Haque, 2021; Pervin, 2025). Criminal "opportunity" is maximized through social engineering rather than through advanced technical hacking since psychological manipulation has a high success rate. Digital anonymity and the perception of MFS providers as faceless corporations facilitate moral disengagement among perpetrators by rationalization.

4.2. RAT Lens: The Coming Together in the MFS Ecosystem

RAT parts come together well in Bangladesh's setting, making perfect conditions for trickery:

- **Offenders with a Reason:** Highly organized crime groups work with great speed using detailed scripts and tricks of the mind (Rahman, 2023).
- **Good Targets:** A big share of new MFS users do not know about basic security rules, making them very open to social tricks (Sultana et al., 2022). This openness is worsened by weak KYC systems that cannot check National ID cards in real-time, letting tricksters open accounts with fake or stolen names (Huda & Sony, 2025).
- **Lack of a Good Protector:** This is seen as the most important security gap. Even though simple SMS OTP checks are there, they can still be easily changed through SIM swapping and phishing attacks (Mondal & Sarkar, 2025). Major gaps are found in real-time AI-based transaction checks that can flag strange AFT attempts (Khan & Ahmed, 2022) and strong changing multi-factor checks.

4.3. NIST CSF Lens: Gaps in the Control Lifecycle

The NIST CSF Analysis indicates the following significant control imbalances:

- **Protect:** Controls predominantly exist in Protect (e.g., SMS OTP, PINs), while the main authentication mechanism (OTP) is still open to interception, SIM-swapping, and social engineering (Mondal & Sarkar, 2025).
- **Detect:** A major gap. Current systems do not have behavioral analytics, AI-powered anomaly detection, or machine learning to flag suspicious AFT activities in real time (Khan & Ahmed, 2022; Sarker, 2025).
- **Respond:** The other critical gap is that there are no standardized rapid-response protocols for freezing transactions and reversing confirmed fraudulent AFTs within guaranteed timeframes which have led to huge losses for users (Hossain et al., 2022).
- **Identify & Recover:** These functions are underemphasized. Poor asset management and resilience planning leave the ecosystem exposed to new attacks and unable to quickly bounce back from successful breaches.

5. Discussion

The synthesis confirms that OTP bypass fraud requires moving beyond a technical solution and confronting a complex socio-technical challenge. The integrated framework shows that isolated interventions like user education (reducing target suitability in RAT) without corresponding enhancements in guardianship (NIST CSF's Detect function) and KYC foundations are inadequate.

Also, the implementation of technical controls (NIST Protect) without addressing fraudster motivations and rationalizations (FTT) leads to misallocated security resources.

The most significant vulnerability results from the interaction between human factors (RAT's suitable target), procedural-technical gaps (NIST's Detect and Respond), and foundational identity verification weaknesses (KYC)—a vulnerability systematically exploited by motivated offenders (FTT).

6. Research Gaps

This review pointed out some important research gaps that need to be explored further:

1. There are very few empirical studies that quantitatively assess the impact of specific NIST CSF controls in the context of MFS in Bangladesh.
2. The social networks and operational dynamics of fraudster syndicates (the "Motivated Offender" in RAT) have not been sufficiently studied in this region.
3. There is limited research on designing and testing "digital guardianship" tools like context-aware notifications for high-risk transactions for users with low digital literacy.
4. There has not been enough study on the economic effects of OTP bypass fraud concerning financial inclusion goals and the profitability of MFS providers.

7. Proposed Conceptual Model

I propose a multi-layered conceptual model that integrates all three frameworks to close the gaps identified in the synthesis.

7.1. Layer 1: Motive Deterrence (Addressing FTT)

- ❖ **National Awareness Campaigns:** Launch large-scale campaigns that draw attention to the legal repercussions associated with certain activities. This approach aims to raise the perceived risk among potential offenders and counteract their attempts at rationalization.
- ❖ **Strengthened Law Enforcement:** Provide dedicated resources for the establishment of specialized units focused on cybercrime, ensuring enhanced collaboration between different agencies.
- ❖ **Fraudster Network Disruption:** Utilize advanced technologies such as artificial intelligence and data analytics to detect and dismantle the operational frameworks used by criminal syndicates.

7.2. Layer 2: Target Hardening & Guardianship (Addressing RAT)

Target Hardening:

- ❖ **Compulsory Interactive Education:** security tutorials inside the app, in the local language and with examples that fit the context, should be implemented as part of MFS onboarding processes.
- ❖ **Enhanced e-KYC:** Incorporating AI-based identity verification, liveness detection, and biometric authentication to stop fake account creation (Sarker, 2025). Real-time NID

verification should be mandatory with direct access to Bangladesh Bank databases (Huda & Sony, 2025).

Capable Guardian:

- ❖ **AI-based Real-time Monitoring:** Use systems based on Natural Language Processing (NLP) and machine learning for user behavior analysis, device fingerprinting, and transaction patterns to enable flagging and blocking of anomalous AFT activities in real-time as described by Sarker & Tanvir in 2025.
- ❖ **Risk-Based MFA:** Innovative authentication solutions could replace or supplement SMS OTP which is vulnerable. The "what we want" model uses pre-configured transaction parameters as authentication factors; this approach looks promising since it is resistant to phishing attacks and SIM-swapping according to Mondal & Sarkar in 2025.

7.3. Layer 3: Control Lifecycle (NIST CSF Implementation)

- ❖ **Identify & Protect:** Improve asset management and governance frameworks. Retain strong customer authentication augmented with contextual, explicit transaction warnings. Assess blockchain-based payment schemes to enhance the integrity of transactions (Dewi et al., 2024).
- ❖ **Detect:** Use the AI guardianship system from Layer 2 as primary detection tool.
- ❖ **Respond & Recover:** Create standardized rapid-response protocols that require freezing of transactions and reversal of fraudulent AFT within stipulated timeframes.
- ❖ **Regulatory Evolution:** Move from reactive risk-based regulation to an active smart regulation model involving coordinated collaboration between government agencies, industry players, and third-party regulators (Huda & Sony, 2025).

This model assumes that these layers need to work together in order to form a complete and strong protection for the MFS ecosystem, moving beyond the usual separate security methods.

Figure 1 & 2 presents a multi-layered conceptual model that integrates insights from all three frameworks in a synergistic manner to address the gaps identified.

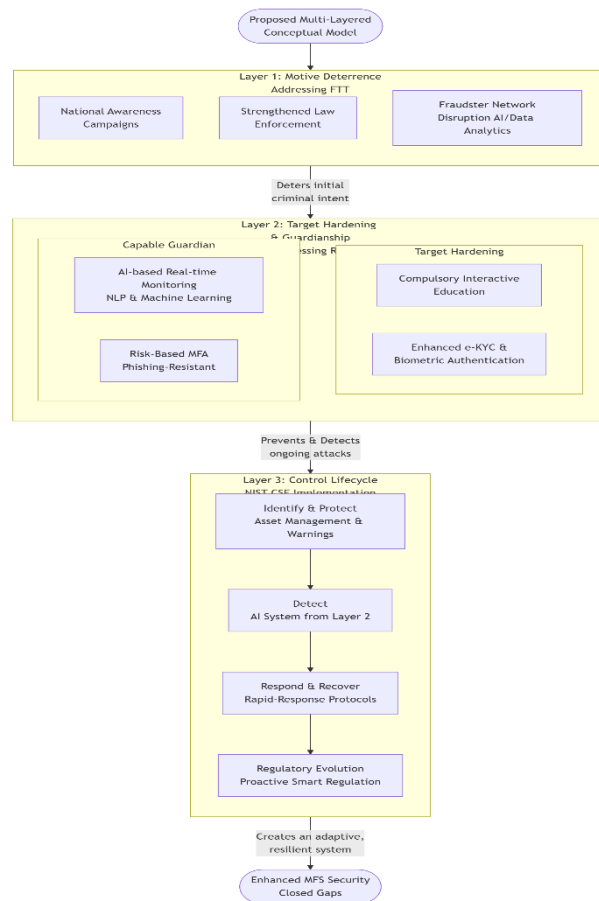


Figure 1: Flowchart.

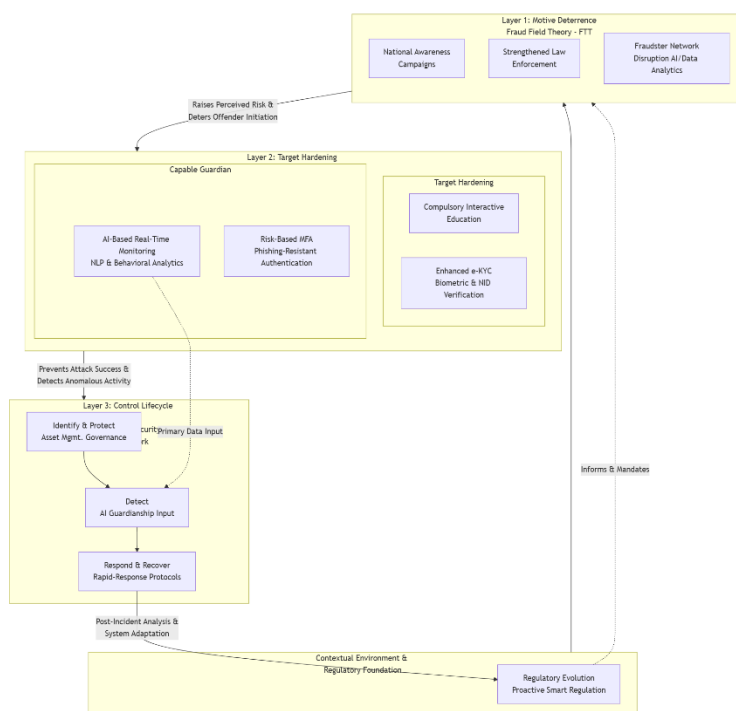


Figure 2: A conceptual model that has many layers and combines ideas from all three frameworks in a way that works together to fill the gaps that have been found.

8. Conclusion

The systematic review reveals that the OTP bypass fraud, driven by social engineering, is a complex socio-technical problem requiring integrated solutions. By bringing together the Fraud Triangle Theory, Routine Activity Theory, and the NIST Cybersecurity Framework, this study offers a multi-dimensional analytical framework for understanding this threat that continues to escalate. This conceptual model serves as an actionable roadmap for stakeholders who want to improve MFS security in Bangladesh.

Findings highlight that technological solutions such as AI-based monitoring and risk-based MFA are necessary but not sufficient if applied separately. Their efficacy is contingent upon simultaneous improvements in public awareness, legal enforcement, basic e-KYC process enhancement, and a shift in regulatory philosophy from risk-based to smart regulation. Subsequent research should focus on empirical validation of this model, culturally relevant AI guardianship tool development for low-digital-literacy users, and an exhaustive study on the economic impacts of OTP fraud. Bangladesh can only strengthen its MFS ecosystem through coordinated efforts among multiple stakeholders to protect the gains made in financial inclusion and ensure a secure digital financial future.

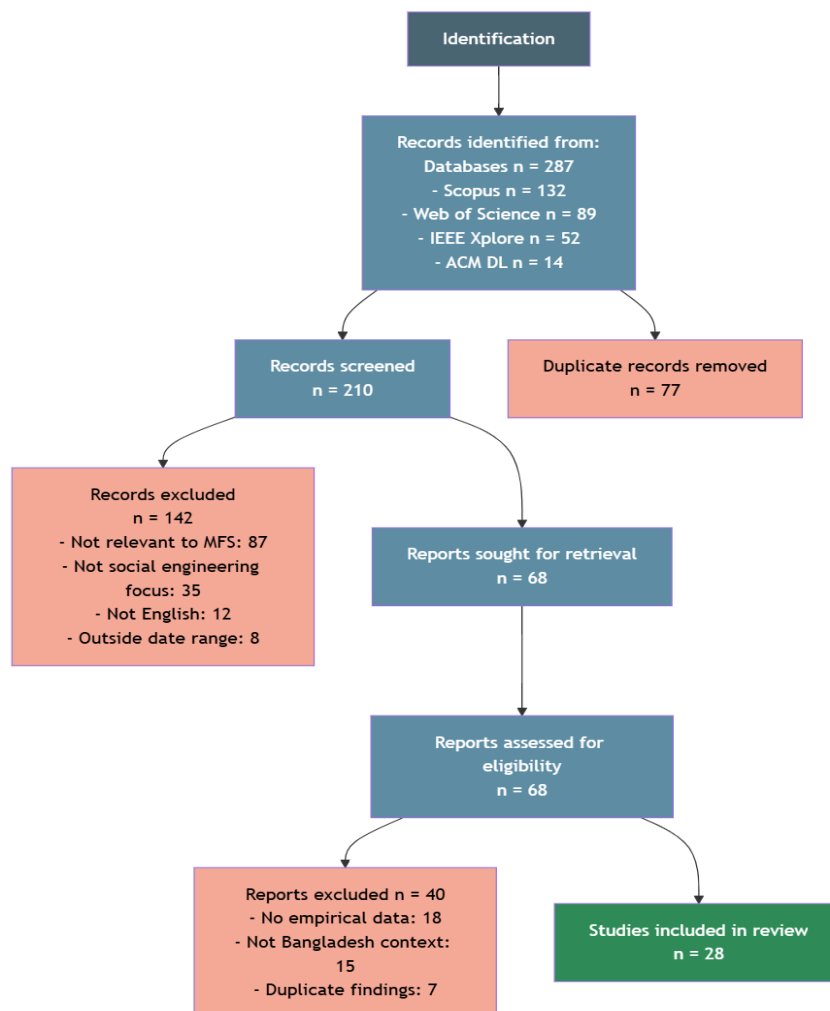
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Appendices

Appendix A: PRISMA Flow Diagram



Reasons for Exclusion at the Title/Abstract Screening Stage (n = 142):

- Not pertinent to MFS security (n = 87): Studies about conventional banking, e-commerce sites, or general cybersecurity without MFS application.
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- Not focused on social engineering (n = 35): Research that deals only with technical attacks (malware or network intrusion) and does not involve social engineering.
-
- Not in English (n = 12): Reports published in Bengali or other languages for which there is no English translation available.
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- Outside the date range (n = 8): Studies published prior to 2015 considered potentially outdated for the rapidly evolving MFS threat landscape.

Exclusions at Full-Text Assessment Stage (n = 40):

- No empirical data (n = 18): Theoretical papers, opinion pieces, or conceptual frameworks that did not include primary data or case studies.
-
- Not specific to Bangladesh context (n = 15): Studies from other geographical contexts with non-transferable findings to Bangladesh's unique MFS ecosystem.
-
- Duplicate findings (n = 7): Multiple publications reporting identical or substantially overlapping datasets.

Characteristics of Included Studies:

The final set of 28 studies achieved exhaustive coverage of:

- Empirical studies on patterns of fraud in MFS from Bangladesh (15)
- Technical analyses regarding vulnerabilities in OTP security (6)
- Policy and regulatory analyses (4)
- User behavior as well as digital literacy studies (3)

Quality Assessment Summary: All included studies were assessed for quality using the Mixed Methods Appraisal Tool (MMAT) 2018. The quality score distribution is as follows: 5 stars - high quality representing nine studies; four stars - good quality representing twelve studies; and three stars - adequate quality comprising seven studies.

No study scoring below the minimum quality threshold of three stars was included in the final synthesis, thereby ensuring that review findings are based on methodologically sound research.

Note: The search strategy was implemented on October 28, 2025; no additional records were found by manually searching reference lists or any other supplementary sources.